

How MODEL-BASED DESIGN And Other Software Toolboxes Can Help You Accelerate EV MOTOR CONTROL DESIGN

A groundbreaking technology that could revolutionise the renewable energy sector, offering new hope for a sustainable future



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A motor controller is a crucial element of an electric vehicle and is increasingly sought to be indigenously developed or sourced by electric 2W and 3W OEMs in India to comply with localisation norms associated with government regulations and incentive schemes. The motor controller firmware can be a differentiating factor for the rider

experience and, as such, offers scope for innovation for startups and established teams alike. This article discusses some key challenges for developers of motor controllers and offers thoughts on mitigating them.

The motor controller is also referred to as a traction 'inverter' since it converts between DC and AC to control the

TABLE 1: TOP-LEVEL COMPARISON OF COMMON SENSOR TYPES USED IN ELECTRIC 2W/3W MOTOR CONTROLLERS

Current Sensor	Precision	Robustness	Cost	Isolation
Hall	Lower	Higher	Higher	Galvanic isolation
Shunt	Higher	Lower (drift, temperature sensitivity)	Lower	No isolation
Position Sensor	Precision	Robustness	Cost	Space
Resolver	Higher	Lower	Higher	Bigger
Encoder	Lower	Higher	Lower	Smaller
Hall	Lower	Higher	Lower	Smaller

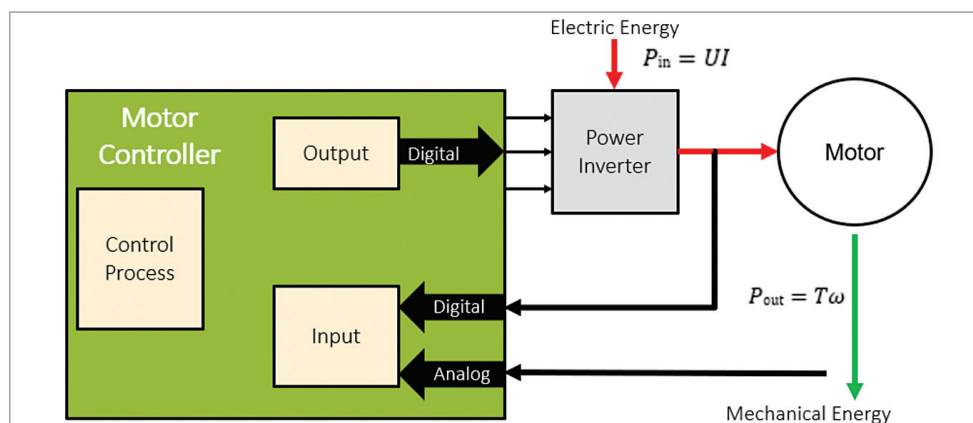


Fig. 1: A simple view of a typical electric 2W/3W motor controller